

Excel Basics Assignment

Q1. Explain the difference between Relative, Absolute and mixed cell referencing.
you have the following dataset:

Vehicle	Price
Car	5,00,000
Bicycle	10,500
Auto-Rickshaw	25,000

Ans:

Relative Cell Referencing : In relative referencing, the cell reference changes automatically when you copy the formula to another cell.

Example: Let column A be Vehicle and Column B be Price so if we calculate 10% tax on vehicle then we will use B2*10% in C1 and by cell reference it will automatically get B3*10% in C2.

Absolute Cell Referencing : In absolute referencing, the cell reference does NOT change when copied.

Example: If we take the reference of the same table so if we enter a new column D1 and In it there is Showroom commission we will write the formula as B2*\$D\$1

Mixed Cell Referencing : In mixed referencing, only row OR column is fixed, not both.

There are two types:

- \$A1 → Column fixed, row changes
- A\$1 → Row fixed, column changes

Example: If we use =B2*D\$1 by same table reference so in this Row 1 is fixed, but column can change if dragged sideways.

Q2. Write a formula to calculate the total sales of car and bicycle only.

Ans: The question is incomplete as No. of unit sold is not mentioned hence, we are not able to accurately get the answer for the total sales but if we take price as sales amount then we can write =SUM (B2:B3) as per Q1 table reference (taking Price as Column B)

Q3. Using the data below, write formula to calculate the average sales of items priced above 100 but less than 300:

Item	Price	Sales
Item A	90	1,000
Item B	150	1,200
Item C	250	1,500

Ans: The formula to calculate the average sales of items priced above 100 but less than 300 with reference of given table is =AVERAGEIFS (C2:C4, B2:B4, ">100", B2:B4, "<300") which gives the answer as 1350

Table Reference for the below given questions are –

OrderID	CustomerName	Region	Product	Quantity	UnitPrice	OrderDate
1001	Amit Sharma	North	Notebook	10	50	05-01-2024
1002	Neha Verma	South	Pen	20	10	06-01-2024
1003	Rahul Singh	East	Marker	15	25	07-01-2024
1004	Pooja Mehta	West	Notebook	5	50	08-01-2024
1005	Ankit Jain	North	Pencil	30	5	10-01-2024
1006	Sneha Patel	South	Notebook	12	50	11-01-2024
1007	Vikas Gupta	East	Eraser	25	8	13-01-2024
1008	Meera Nair	West	Pen	18	10	15-01-2024
1009	Anjali Rao	North	Marker	22	25	17-01-2024
1010	Manoj Das	South	Notebook	8	50	18-01-2024
1011	Ritika Sen	East	Pencil	40	5	19-01-2024
1012	Karan Malhotra	West	Eraser	35	8	20-01-2024
1013	Aryan Joshi	North	Notebook	14	50	22-01-2024
1014	Divya Kapoor	South	Pen	28	10	23-01-2024
1015	Sahil Arora	East	Marker	12	25	24-01-2024
1016	Muskan Jain	West	Pencil	25	5	26-01-2024
1017	Rohan Gupta	North	Eraser	20	8	28-01-2024
1018	Nidhi Sharma	South	Notebook	10	50	29-01-2024
1019	Tarun Verma	East	Pen	15	10	30-01-2024
1020	Simran Kaur	West	Marker	18	25	01-02-2024

1021	Deepak Rao	North	Notebook	9	50	02-02-2024
1022	Payal Mehta	South	Pencil	35	5	03-02-2024
1023	Aditya Singh	East	Eraser	30	8	04-02-2024
1024	Shreya Nair	West	Notebook	11	50	06-02-2024
1025	Sanjay Kumar	North	Pen	14	10	07-02-2024
1026	Aarti Sharma	South	Marker	16	25	08-02-2024
1027	Rajesh Patel	East	Pencil	22	5	09-02-2024
1028	Preeti Joshi	West	Eraser	27	8	10-02-2024
1029	Anurag Mishra	North	Notebook	13	50	12-02-2024
1030	Sanya Kapoor	South	Pen	32	10	13-02-2024
1031	Harsh Yadav	East	Marker	10	25	14-02-2024
1032	Kajal Rani	West	Pencil	18	5	16-02-2024
1033	Varun Sharma	North	Eraser	22	8	17-02-2024
1034	Swati Arora	South	Notebook	17	50	18-02-2024
1035	Akhil Jain	East	Pen	19	10	19-02-2024
1036	Priyanka Das	West	Marker	15	25	21-02-2024
1037	Ravi Nair	North	Pencil	28	5	22-02-2024
1038	Komal Singh	South	Eraser	24	8	23-02-2024
1039	Ayesha Khan	East	Notebook	16	50	25-02-2024
1040	Arjun Mehta	West	Pen	27	10	26-02-2024
1041	Sonali Gupta	North	Marker	21	25	27-02-2024
1042	Neeraj Kapoor	South	Pencil	30	5	28-02-2024
1043	Shruti Sharma	East	Eraser	26	8	01-03-2024
1044	Yash Arora	West	Notebook	14	50	02-03-2024
1045	Manish Das	North	Pen	20	10	03-03-2024
1046	Rupal Nair	South	Marker	17	25	04-03-2024
1047	Sonia Verma	East	Pencil	19	5	05-03-2024
1048	Kunal Singh	West	Eraser	32	8	06-03-2024
1049	Ritika Jain	North	Notebook	12	50	07-03-2024
1050	Vivek Malhotra	South	Pen	25	10	08-03-2024

Q4. Count how many customer names are recorded.

Ans: The Formula to calculate how many customer names we have in this record is
=COUNTA(Table1[CustomerName]) if counts the cells which are not empty
For practical use have a look on attached excel sheet

Q5. Calculate the total sales for each row using a formula

Ans: We can use the formula $=[@Quantity]*[@UnitPrice]$ to get Total Sales
For Practical answer check attached excel sheet

Q6. Calculate the total sales of notebooks in north region only.

Ans: we can use the formula
 $=SUMPRODUCT((C2:C51="North")*(D2:D51="Notebook")*(E2:E51)*(F2:F51))$

Q7. Create a column chart showing total sales by product.

Ans: Check attached Excel Sheet

Q8. Insert a line chart showing daily sales trend.

Ans: Check attached Excel Sheet